The Distance Route: split recovery from a sub-sampled double-centred distance matrix

Abstract

Let $\mathcal{D}$ be the matrix of path distances between the m leaves of a weighted tree and let $\mathcal{B} = H\mathcal{D}H$ be its double centring, $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$. We give a self-contained account of why the sign pattern of the leading-magnitude eigenvector of $\mathcal{B}$ is a bipartition of the leaves cut by a single edge of the tree, and of how much of the matrix may be discarded before that sign pattern changes. Two exact structure theorems carry the argument. The first writes $\mathcal{B}$ as a negatively weighted sum of rank-one split projectors, $\mathcal{B} = -\sum_e w_e\, \hat{e}_e\hat{e}_e^\top$ with $w_e = 2\ell_e a_e b_e/m$; the second identifies $-\frac{1}{2}\mathcal{B}$ with the Moore–Penrose pseudoinverse of the tree Laplacian Schur-complemented onto the leaves. The first supplies the spectral gap, and an explicit preference for balanced bipartitions that the Laplacian operator does not share; the second supplies validity, with no quantitative hypothesis at all. Against these we run a sampling model in which each pair is observed independently with probability p: an operator-norm bound for the inverse-probability estimator, a Davis–Kahan step, and a one-step refinement whose per-leaf statistic concentrates by a scalar Bernstein inequality. The conclusion is that $p = \Theta(\log m/m)$ suffices to reproduce exactly the partition the fully observed operator returns.

1 Setting

Let T be a tree with leaf set X, $|X| = m \geq 4$, internal vertex set R, edge set E, and a strictly positive length ℓ_e on each edge. For $i, j \in X$ let $\mathcal{D}_{ij}$ be the total length of the unique i–j path, and write

$$d_{\max} := \max_{i,j\in X} \mathcal{D}_{ij}. \tag{1}$$

Let $H := I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$ be the centring projector and set

$$\mathcal{B} := H\,\mathcal{D}\,H, \qquad \mathcal{M} := -\mathcal{B}. \tag{2}$$

The object of study is the *sign rule*: take $u^{(1)}$, a unit eigenvector of $\mathcal{B}$ belonging to the eigenvalue of largest absolute value, and partition X by $\operatorname{sign}(u^{(1)})$.

Remark 1 (Sign convention). Corollary 1 below shows $\mathcal{B} \preceq 0$, so the eigenvalue of largest absolute value is the *most negative* one, and $u^{(1)}$ is the top eigenvector of $\mathcal{M} \succeq 0$. The literature that uses this operator describes $u^{(1)}$ as "leading" [3]; on $\mathcal{B}$ that word must be read as largest in magnitude, not largest algebraically, since the largest algebraic eigenvalue of $\mathcal{B}$ is the eigenvalue 0 carried by $\mathbf{1}$.

Throughout, $A_e \subset X$ and $B_e = X \setminus A_e$ denote the two leaf sets obtained by deleting edge e from T, with $a_e = |A_e|$ and $b_e = |B_e|$; both are non-empty. A bipartition of X of this form is called a *split of* T. For $A \subseteq X$ we write $\mathbf{1}_A$ for its indicator vector.

2 The split-covariance identity

The first structure theorem needs nothing beyond the definition of an additive metric and one line of algebra.

Lemma 1 (Path decomposition)**.** *For $e \in E$ put $S_e := \mathbf{1}_{A_e}\mathbf{1}_{B_e}^\top + \mathbf{1}_{B_e}\mathbf{1}_{A_e}^\top$. Then*

$$\mathcal{D} = \sum_{e\in E} \ell_e\, S_e. \tag{3}$$

Proof. T is a tree, so the i–j path is unique and $\mathcal{D}_{ij} = \sum_{e\in E} \ell_e\, \mathbf{1}\{e \text{ lies on the } i\text{–}j \text{ path}\}$. Deleting e splits T into the two components containing A_e and B_e respectively, and e lies on the i–j path exactly when i and j fall in different components, i.e. exactly when $(S_e)_{ij} = 1$. □

Lemma 2 (Rank-one centring)**.** *Let $A \subseteq X$ with $a = |A| \in \{1, \dots, m-1\}$, $b = m - a$, and set $\tilde{u} := H\mathbf{1}_A$. Then*

$$H\big(\mathbf{1}_A\mathbf{1}_B^\top + \mathbf{1}_B\mathbf{1}_A^\top\big)H = -2\,\tilde{u}\tilde{u}^\top, \qquad \|\tilde{u}\|^2 = \frac{ab}{m}, \tag{4}$$

where $B = X \setminus A$. Moreover $\tilde{u}_i = b/m$ for $i \in A$ and $\tilde{u}_i = -a/m$ for $i \in B$, so with $\hat{e} := \tilde{u}/\|\tilde{u}\|$,

$$\hat{e}_i = \sqrt{\tfrac{b}{am}} \;\; (i \in A), \qquad \hat{e}_i = -\sqrt{\tfrac{a}{bm}} \;\; (i \in B), \qquad \min_i |\hat{e}_i| = \sqrt{\frac{\min(a,b)}{m\,\max(a,b)}}. \tag{5}$$

Proof. $H\mathbf{1} = 0$ and $\mathbf{1}_B = \mathbf{1} - \mathbf{1}_A$, hence $H\mathbf{1}_B = -\tilde{u}$. Therefore $H(\mathbf{1}_A\mathbf{1}_B^\top + \mathbf{1}_B\mathbf{1}_A^\top)H = (H\mathbf{1}_A)(H\mathbf{1}_B)^\top + (H\mathbf{1}_B)(H\mathbf{1}_A)^\top = -2\tilde{u}\tilde{u}^\top$. The entries of $\tilde{u} = \mathbf{1}_A - \frac{a}{m}\mathbf{1}$ are as stated, so $\|\tilde{u}\|^2 = a(b/m)^2 + b(a/m)^2 = ab(a+b)/m^2 = ab/m$, and dividing gives (5). □

Write $\tilde{u}_e := H\mathbf{1}_{A_e}$ and $\hat{e}_e := \tilde{u}_e/\|\tilde{u}_e\|$. Note that $\operatorname{sign}(\hat{e}_e)$ *is* the split (A_e, B_e), with no zero coordinate.

Theorem 1 (Split covariance)**.**

$$\mathcal{B} = -\sum_{e\in E} w_e\, \hat{e}_e\hat{e}_e^\top, \qquad w_e := \frac{2\,\ell_e\, a_e\, b_e}{m} \;>\; 0. \tag{6}$$

Proof. Apply $H(\cdot)H$ to (3) and use (4) term by term: $H(\ell_e S_e)H = -2\ell_e\tilde{u}_e\tilde{u}_e^\top = -2\ell_e\,\|\tilde{u}_e\|^2\,\hat{e}_e\hat{e}_e^\top = -w_e\hat{e}_e\hat{e}_e^\top$. □

Theorem 1 is the exact and finite counterpart, for a tree metric, of a canonical split decomposition [5]: the split system of T is its edge set, so no limiting or canonical-form argument is required.

Corollary 1 (Definiteness, rank, trace)**.** *$\mathcal{B} \preceq 0$ with $\mathcal{B}\mathbf{1} = 0$; $\operatorname{rank}\mathcal{B} = m - 1$, so the inertia of $\mathcal{B}$ is $(0, 1, m-1)$; and*

$$\sum_{e\in E} w_e = \operatorname{tr}\mathcal{M} = \frac{1}{m}\,\mathbf{1}^\top\mathcal{D}\,\mathbf{1}. \tag{7}$$

Proof. Negative semidefiniteness and $\mathcal{B}\mathbf{1} = 0$ are immediate from (6) and $\hat{e}_e^\top\mathbf{1} = 0$. For the rank: the m pendant edges contribute the vectors $\tilde{u}_{\{i\}} = He_i$, $i \in X$, which span $\mathbf{1}^\perp$, a space of dimension $m-1$; since every $w_e > 0$ the quadratic form $x^\top\mathcal{M}x = \sum_e w_e\langle\hat{e}_e, x\rangle^2$ vanishes on $\mathbf{1}^\perp$ only at 0. For the trace, each $\hat{e}_e$ is a unit vector, so $\operatorname{tr}\mathcal{M} = \sum_e w_e$; independently $\operatorname{tr}(H\mathcal{D}H) = \operatorname{tr}(\mathcal{D}H) = \operatorname{tr}\mathcal{D} - \frac{1}{m}\mathbf{1}^\top\mathcal{D}\mathbf{1} = -\frac{1}{m}\mathbf{1}^\top\mathcal{D}\mathbf{1}$ because $\mathcal{D}$ has zero diagonal. □

Remark 2 (Inertia of $\mathcal{D}$ itself). Corollary 1 is consistent with, and is the centred form of, the classical statement that the distance matrix of a tree on m vertices has exactly one positive and $m-1$ negative eigenvalues. For *weighted* trees — the case here — the citable statement is the $s = 1$ specialisation of Bapat's matrix-weight determinant theorem [6]; Graham and Pollak proved the determinant $\det \mathcal{D} = (-1)^{m-1}(m-1)2^{m-2}$ for the unweighted case [19], and Graham and Lovász the associated polynomials [18]. The wider family of exact relations between the distance matrix of a tree and its Laplacian, including the closed form of $\mathcal{D}^{-1}$, is collected in [8, 7].

Remark 3 ($\mathcal{B}$ is the classical scaling operator). $G = -\frac{1}{2}\mathcal{B}$ is the Gram matrix of classical multidimensional scaling applied to $\mathcal{D}$ [28, 26, 16], so $\operatorname{sign}(u^{(1)})$ is the sign of the first principal coordinate of the leaves. Corollary 1 states that this Gram matrix is positive semidefinite for every weighted tree, which is the assertion that the tree metric $\mathcal{D}$ — not its square — is a squared Euclidean distance, i.e. that $\sqrt{\mathcal{D}}$ embeds isometrically in $\mathbb{R}^{m-1}$ [24, 17]; that phylogenetic distance matrices have this property, and the elementary reason for it, are discussed in [12, 22]. The geometry of where the principal axes cut a tree is treated in [14].

Lemma 3 (Gap bracket)**.** $\max_{e\in E} w_e \;\le\; \lambda_1(\mathcal{M}) \;\le\; \sum_{e\in E} w_e.$

Proof. Lower: $\lambda_1(\mathcal{M}) \ge \hat{e}_e^\top \mathcal{M}\, \hat{e}_e \ge w_e$ for each e, since every summand in (6) is positive semidefinite. Upper: $\lambda_1(\mathcal{M}) \le \operatorname{tr} \mathcal{M}$, again by positive semidefiniteness. □

The inner products between split directions are also closed-form. Any two distinct edges of a tree induce nested splits, so after orienting A_f appropriately exactly one of the two cases below applies.

Lemma 4 (Split Gram matrix)**.** *Let $e, f \in E$, $e \ne f$. Then*

$$\langle \tilde{u}_e, \tilde{u}_f \rangle = \begin{cases} \dfrac{a_f\, b_e}{m}, & A_f \subseteq A_e, \\ -\dfrac{a_e\, a_f}{m}, & A_f \subseteq B_e. \end{cases} \tag{8}$$

Proof. $\langle H\mathbf{1}_{A_e}, H\mathbf{1}_{A_f}\rangle = \mathbf{1}_{A_e}^\top H\mathbf{1}_{A_f} = |A_e \cap A_f| - a_e a_f/m$. In the first case $|A_e \cap A_f| = a_f$, giving $a_f(1 - a_e/m) = a_f b_e/m$; in the second $|A_e \cap A_f| = 0$. □

Equation (8) is never zero, which is the reason the sign rule does not simply return the edge of largest weight w_e: the directions $\hat{e}_e$ are not orthogonal, and the top eigenvector of $\mathcal{M}$ responds to the whole correlated family. Section 8 records how often the two differ.

3 Validity, unconditionally

The second structure theorem says that $-\frac{1}{2}\mathcal{B}$ is not merely some positive semidefinite matrix but a Laplacian pseudoinverse. It is asserted without proof in [20, §4.4, §4.8.1]; we prove it, because everything in this section rests on it.

Give each edge the conductance $c_e := 1/\ell_e$ and let $L \in \mathbb{R}^{V\times V}$, $V = X \sqcup R$, be the weighted Laplacian of T. Order V with the leaves first,

$$L = \begin{pmatrix} L_{XX} & L_{XR} \\ L_{RX} & L_{RR} \end{pmatrix}, \qquad L^* := L/L_{RR} = L_{XX} - L_{XR}L_{RR}^{-1}L_{RX}. \tag{9}$$

L_{RR} is a principal submatrix of a connected graph Laplacian obtained by deleting a non-empty vertex set, hence positive definite, so L^* is well defined.

Theorem 2 (Gower matrix as a Laplacian pseudoinverse)**.**

$$G := -\tfrac{1}{2}\, H\mathcal{D}H = \big(L^*\big)^{\dagger}. \tag{10}$$

Proof. Step 1: L^* is the boundary current-to-potential map. Let $x \in \mathbb{R}^X$ with $\mathbf{1}^\top x = 0$, and let $v = (v_X, v_R)$ solve $Lv = (x, 0)$. The second block gives $v_R = -L_{RR}^{-1} L_{RX} v_X$, and substituting into the first gives $L^* v_X = x$. So injecting the current x at the leaves and none at the internal vertices produces leaf potentials v_X related to x by L^*. Consequently the effective resistance between $i, j \in X$ in the network (T, c) equals $(e_i - e_j)^\top (L^*)^\dagger (e_i - e_j)$.

Step 2: effective resistance is the tree metric. In a tree the i–j path is unique, so resistors in series give effective resistance $\sum_{e \in \text{path}} 1/c_e = \sum_{e\in\text{path}} \ell_e = \mathcal{D}_{ij}$. Hence $\mathcal{D}$ is exactly the resistance matrix of L^*.

Step 3: double centring inverts the resistance formula. For any symmetric positive semidefinite L^* with kernel $\mathbf{1}$, write $g := \operatorname{diag}\big((L^*)^\dagger\big)$. Expanding Step 1, $\mathcal{D}_{ij} = g_i + g_j - 2(L^*)^\dagger_{ij}$, i.e. $\mathcal{D} = \mathbf{1}g^\top + g\mathbf{1}^\top - 2(L^*)^\dagger$. Apply $H(\cdot)H$: the first two terms are annihilated because $H\mathbf{1} = 0$, so $H\mathcal{D}H = -2\, H(L^*)^\dagger H$. Finally $(L^*)^\dagger \mathbf{1} = 0$ and $(L^*)^\dagger$ is symmetric, so $H(L^*)^\dagger H = (L^*)^\dagger$, which is (10). □

Pseudoinversion fixes eigenvectors and inverts the non-zero eigenvalues. Both G and L^* have kernel exactly span($\mathbf{1}$) (Corollary 1 for G; connectivity for L^*), so:

Corollary 2 (Spectral correspondence)**.** *G and L^* have the same eigenvectors. The eigenvector of G at its largest eigenvalue — equivalently, by Remark 1, the vector $u^{(1)}$ of Section 1 — is the Fiedler vector of L^*, and $\lambda_{\max}(G) = 1/\lambda_2(L^*)$.*

L^* is the Laplacian of a weighted graph on the leaves obtained from T by eliminating internal vertices, and the sign pattern of its Fiedler vector is governed by a theorem of Stone and Griffing [25, Thm. 3.3]: it partitions X into V_+ and V_- such that, together with a suitable partition of the internal vertices, both parts induce connected subgraphs of T. This is the Schur-complemented counterpart of Fiedler's classical result that each nodal domain of a Fiedler vector induces a connected subgraph [13]; the behaviour of the later eigenvectors under the same reduction is described in [21]. A bipartition of the leaves whose two sides induce connected subtrees is a split of T. Hence:

Corollary 3 (Validity of the sign rule)**.** sign($u^{(1)}$) *is the bipartition (A_e, B_e) induced by some single edge $e \in E$, and $u^{(1)}$ has no zero coordinate. No quantitative hypothesis on T is required.*

Definition 1 (Dominant edge)**.** By Corollary 3 there is an edge $e_* \in E$ with $\operatorname{sign}(u^{(1)}) = (A_{e_*}, B_{e_*})$. Write $A_* := A_{e_*}$, $B_* := B_{e_*}$, $a := |A_*|$, $b := |B_*|$, $\ell_* := \ell_{e_*}$, $w_* := w_{e_*} = 2\ell_* ab/m$, and $\delta_* := \min_i |u_i^{(1)}| > 0$.

This is the target of everything that follows. The task of the remaining sections is not to identify e_* — Corollary 3 has already guaranteed that the exact rule returns a split of T — but to say how much of $\mathcal{D}$ may be discarded before the sampled rule stops returning (A_*, B_*).

Remark 4 (Relation to the Fiedler vector of a similarity Laplacian)*.* Corollary 3 certifies that the distance operator returns *a* split. It does not assert that this split is the one a Laplacian built from a similarity matrix returns, and in general it is not: the two operators are different functions of the tree and they select different edges. The quantitative reason is isolated in Proposition 2.

4 Population landmarks

Three population quantities enter the sampling analysis. Each is a property of the fully observed operator alone.

Assumption D1 (Spectral gap)**.** $\lambda_1(\mathcal{M})$ is simple, and $\Delta := \lambda_1(\mathcal{M}) - \lambda_2(\mathcal{M}) \;\geq\; \gamma\, w_*$ for a constant $\gamma > 0$.

Assumption D2 (Delocalisation)**.** $\delta_* = \min_i |u_i^{(1)}| \;\geq\; c_\delta/\sqrt{m}$ for a constant $c_\delta > 0$.

Assumption D3 (Row separation)**.** For every $i \in X$, with the convention that the sums omit $j = i$,

$$s_i := \left| \frac{1}{|B_*|} \sum_{j \in B_*} \mathcal{D}_{ij} - \frac{1}{|A_*| - 1} \sum_{j \in A_*} \mathcal{D}_{ij} \right| \;\geq\; s_0 > 0 \quad (i \in A_*), \tag{11}$$

and symmetrically for $i \in B_*$. Write $\kappa_{\mathrm{ref}} := d_{\max}/s_0$.

Assumption D2 is a normalisation, not a restriction on the tree: by (5) the split direction $\hat{e}_{e_*}$ itself has $\min_i |\hat{e}_i| = \sqrt{\min(a,b)/(m \max(a,b))}$, which is $c_\delta/\sqrt{m}$ with $c_\delta = (1+\eta)^{-1/2}$ at imbalance $\eta = \max(a,b)/\min(a,b)$. Assumption D3 says the same split is visible in a single row of $\mathcal{D}$; by Lemma 1 the edge e_* alone contributes exactly ℓ_* to s_i, so s_0 is the part of that contribution that survives the other edges.

A family in which all three are exact. Let T be the balanced binary tree on $m = 2^k$ leaves in which the two edges at the root have length L and every other edge has length g, so that $\mathcal{D}_{ij} = 2L + 2g(k-1)$ across the root split and $2gd$ within, d being the depth of the meet. Then A_*, B_* are the two halves, $w_* = 2 \cdot (2L) \cdot \frac{m}{2} \cdot \frac{m}{2}/m = Lm$, and $u^{(1)}$ equals $\hat{e}_{e_*}$ exactly, so $\delta_* = 1/\sqrt{m}$ and $c_\delta = 1$. Numerically, for $g = 1$:

m	L	$\lvert\lambda_1(\mathcal{M})\rvert$	$\lvert\lambda_2(\mathcal{M})\rvert$	Δ	$w_* = Lm$	$\sqrt{m}\,\delta_*$
64	0.5	94.00	62.00	32.00	32.00	1.000
64	1.0	126.00	62.00	64.00	64.00	1.000
64	2.0	190.00	62.00	128.00	128.00	1.000
64	4.0	318.00	62.00	256.00	256.00	1.000
256	0.5	382.00	254.00	128.00	128.00	1.000
256	1.0	510.00	254.00	256.00	256.00	1.000
256	2.0	766.00	254.00	512.00	512.00	1.000
256	4.0	1278.00	254.00	1024.00	1024.00	1.000

so $\Delta = w_*$ identically and Assumption D1 holds with $\gamma = 1$, while $\lambda_2(\mathcal{M}) = m - 2$ is set by the internal edges alone. Here $d_{\max} = 2L + 2g(k-1)$, and the two ratios that govern the rate are $\kappa := d_{\max}/\ell_*$ and $\kappa_{\mathrm{ref}} = d_{\max}/s_0$.

5 Observation model and spectral noise

Each unordered pair is observed independently with probability p: let $\Omega_{ij} = \Omega_{ji} \sim \mathrm{Bernoulli}(p)$ for $i < j$, independent across pairs, $\Omega_{ii} = 0$, and form the inverse-probability estimator

$$\widehat{\mathcal{D}} := p^{-1}\, \mathcal{D} \circ \Omega, \qquad \widehat{\mathcal{B}} := H\, \widehat{\mathcal{D}}\, H. \tag{12}$$

Lemma 5 (Unbiasedness)**.** *$\mathbb{E}\widehat{\mathcal{D}} = \mathcal{D}$ and $\mathbb{E}\widehat{\mathcal{B}} = \mathcal{B}$. Moreover, with $Z := \widehat{\mathcal{D}} - \mathcal{D}$,*

$$\left\|\widehat{\mathcal{B}} - \mathcal{B}\right\|_2 = \|HZH\|_2 \leq \|Z\|_2 . \tag{13}$$

Proof. $\mathbb{E}\Omega_{ij} = p$ gives the first claim entrywise; H is deterministic, so it passes through the expectation and centring introduces no bias. For (13), H is an orthogonal projector, so $\|H\|_2 = 1$. □

The entries of Z on and above the diagonal are independent and mean zero, with

$$|Z_{ij}| \leq d_{\max}/p, \qquad \mathrm{Var}(Z_{ij}) = \mathcal{D}_{ij}^2 \frac{1-p}{p} \leq \frac{d_{\max}^2}{p}. \tag{14}$$

Lemma 6 (Spectral error)**.** *There are absolute constants C_0, C_1, C_2 such that if $pm \geq C_0$ then*

$$\|Z\|_2 \ \leq\ 2.01\, d_{\max}\sqrt{\frac{m}{p}} \qquad \textit{with probability at least } 1 - C_1 e^{-C_2 pm}. \tag{15}$$

Proof. Z is symmetric with independent, mean-zero entries on and above the diagonal, bounded in absolute value by $d_{\max}/p$ and with variance at most $d_{\max}^2/p$ by (14). This is the hypothesis of [9, Thm. 8.4], applied to the matrix $pZ/d_{\max}$, whose entries are bounded by 1 and have variance at most p. □

Remark 5 (Which concentration inequality, and why)*.* Lemma 6 uses [9] because it is stated for exactly this ensemble: a symmetric matrix with independent entries on and above the diagonal. A fully self-contained derivation is available from the matrix Bernstein inequality [27, Thm. 1.4] — writing Z as a sum of $\binom{m}{2}$ independent symmetric rank-two terms, with scale $d_{\max}/p$ and variance proxy $md_{\max}^2/p$, gives $\|Z\|_2 \lesssim d_{\max}\big(\sqrt{m\log m/p} + \log m/p\big)$, which costs a factor $\sqrt{\log m}$. Two neighbouring results do not apply as stated: [2, Thm. 2] is the algorithmic precedent for the same $\sqrt{m/p}$ rate but is stated for the non-symmetric independent-entry model, and [4, Thm. 1.1] is sharp but Gaussian.

6 From the spectrum to the labels

Let $\hat{u}^{(1)}$ be a unit eigenvector of $\widehat{\mathcal{B}}$ at its largest-magnitude eigenvalue, sign-aligned so that $\langle \hat{u}^{(1)}, u^{(1)}\rangle \geq 0$.

Lemma 7 (ℓ_2 control)**.** *Under Assumption D1, on the event of Lemma 6,*

$$\left\|\hat{u}^{(1)} - u^{(1)}\right\|_2 \ \leq\ \sqrt{2}\,\sin\Theta(\hat{u}^{(1)}, u^{(1)}) \ \leq\ \frac{2\sqrt{2}\ \|Z\|_2}{\Delta} \ \leq\ \frac{5.69\, d_{\max}}{\gamma\, w_*}\sqrt{\frac{m}{p}}. \tag{16}$$

In particular, if A_* and B_* are balanced up to a constant, so that $w_* \asymp \ell_* m$, then with $\kappa = d_{\max}/\ell_*$,

$$\left\|\hat{u}^{(1)} - u^{(1)}\right\|_2 \ \lesssim\ \frac{\kappa}{\gamma\sqrt{pm}}. \tag{17}$$

Proof. Apply the one-dimensional case of the statistical variant [29, Cor. 1] of the Davis–Kahan $\sin\Theta$ theorem [11] to the symmetric pair $\mathcal{M}$, $-\widehat{\mathcal{B}}$, whose difference has operator norm at most $\|Z\|_2$ by (13); the corollary requires only the *population* gap Δ, which is Assumption D1. The bound $\|\hat{u}^{(1)} - u^{(1)}\|_2 \leq \sqrt{2}\sin\Theta$ holds after sign alignment. Substitute (15) and $\Delta \geq \gamma w_*$. □

Proposition 1 (Misclassification count). *Let $\mathcal{E} := \{i : \operatorname{sign}(\hat{u}_i^{(1)}) \neq \operatorname{sign}(u_i^{(1)})\}$. Under Assumptions D1 and D2,*

$$\frac{|\mathcal{E}|}{m} \;\le\; \frac{\left\|\hat{u}^{(1)} - u^{(1)}\right\|_2^2}{m\,\delta_*^2} \;\le\; \frac{1}{c_\delta^2}\left\|\hat{u}^{(1)} - u^{(1)}\right\|_2^2 \;\lesssim\; \frac{\kappa^2}{c_\delta^2\gamma^2\,pm}. \tag{18}$$

Proof. If $i \in \mathcal{E}$ then $\hat{u}_i^{(1)}$ and $u_i^{(1)}$ have opposite signs, so $|\hat{u}_i^{(1)} - u_i^{(1)}| \ge |u_i^{(1)}| \ge \delta_*$. Hence $|\mathcal{E}|\,\delta_*^2 \le \left\|\hat{u}^{(1)} - u^{(1)}\right\|_2^2$. Insert Assumption D2 and (17). □

Remark 6 (Why not carry the eigenvector to ℓ_∞ directly). Exact recovery by the raw sign rule needs $\left\|\hat{u}^{(1)} - u^{(1)}\right\|_\infty < \delta_*$. The only dimension-free transfer, $\|\cdot\|_\infty \le \|\cdot\|_2$, charges every leaf as though it were the worst one, and against $\delta_* \asymp m^{-1/2}$ that costs a factor m: it turns (17) into the requirement $p \gtrsim \kappa^2/(c_\delta\gamma)^2$, a constant sampling rate. Section 7 recovers the $\log m/m$ rate by adding a second stage instead; Remark 9 names the alternative.

7 Exact recovery by one-step refinement

Let $(\widehat{C}_0, \widehat{C}_1)$ be the partition $\operatorname{sign}(\hat{u}^{(1)})$ from Section 6, and define the refined labelling: assign i to the part c minimising

$$\psi_c(i) := \frac{1}{|\widehat{C}_c \setminus \{i\}|}\sum_{j \in \widehat{C}_c\setminus\{i\}} \widehat{\mathcal{D}}_{ij}, \qquad c \in \{0, 1\}. \tag{19}$$

Lemma 8 (Per-leaf concentration). *Fix i and a set $C \subseteq X\setminus\{i\}$ with $|C| \ge m/4$. For any $\tau \in (0,1)$,*

$$\mathbb{P}\Big(\big|\psi_C(i) - \bar{\mathcal{D}}_{iC}\big| \ge d_{\max}\sqrt{\frac{8\log(m/\tau)}{p\,|C|}} + \frac{4d_{\max}\log(m/\tau)}{3p\,|C|}\Big) \le \frac{2\tau}{m}, \tag{20}$$

where $\bar{\mathcal{D}}_{iC} := |C|^{-1}\sum_{j\in C}\mathcal{D}_{ij}$.

Proof. $\psi_C(i) = |C|^{-1}\sum_{j\in C} p^{-1}\Omega_{ij}\mathcal{D}_{ij}$ is an average of $|C|$ independent random variables with mean $\bar{\mathcal{D}}_{iC}$, each bounded in absolute value by $d_{\max}/(p|C|)$ after scaling and with variance at most $d_{\max}^2/(p|C|^2)$ by (14). Bernstein's inequality for bounded independent summands gives (20). □

Theorem 3 (Exact recovery). *Assume Assumptions D1 to D3, and that the imbalance $\eta = \max(a,b)/\min(a,b)$ is bounded. There are constants $C = C(\gamma, c_\delta, \eta)$ and $c > 0$ such that if*

$$p \;\ge\; C\left(\kappa_{\mathrm{ref}}^2 + \kappa^2\right)\frac{\log m}{m}, \qquad \kappa = \frac{d_{\max}}{\ell_*}, \quad \kappa_{\mathrm{ref}} = \frac{d_{\max}}{s_0}, \tag{21}$$

then with probability at least $1 - m^{-c}$ the refined labelling equals $\operatorname{sign}(u^{(1)})$ at every leaf. By Corollary 3 that labelling is the bipartition of X induced by the edge e_* of T.

Proof. Write $\varepsilon := |\mathcal{E}|/m$ for the stage-one error rate. By Proposition 1 and (21), $\varepsilon \lesssim \kappa^2/(c_\delta\gamma)^2 pm \le s_0/(8d_{\max})$ once C is large enough; in particular $\varepsilon < \frac{1}{2}\min(a,b)/m$, so both $\widehat{C}_0$ and $\widehat{C}_1$ have at least $m/4$ elements and each is within εm leaves of the corresponding true part.

Fix $i \in A_*$. Replacing $\widehat{C}_0, \widehat{C}_1$ by A_*, B_* inside (19) changes each average by at most $2\varepsilon d_{\max}\cdot(m/\min|\widehat{C}_c|) \le 8\varepsilon d_{\max} \le s_0/4$, because at most εm summands differ and each is bounded by $d_{\max}$. By Assumption D3 the population difference $\bar{\mathcal{D}}_{iB_*} - \bar{\mathcal{D}}_{iA_*}$ has absolute value at least s_0 and the sign that identifies i as belonging to A_*. It therefore suffices that the stochastic deviation of each of the two averages be below $s_0/4$. By Lemma 8 with $\tau = m^{-c}$ and $|C| \geq m/4$, both deviations are at most

$$d_{\max}\sqrt{\frac{32(1+c)\log m}{p\,m}} + \frac{16(1+c)d_{\max}\log m}{3pm} \;\leq\; \frac{s_0}{4}$$

whenever $pm \geq C'\kappa_{\mathrm{ref}}^2 \log m$, which is (21). A union bound over the $2m$ statistics $\psi_c(i)$ costs a factor $2m$, absorbed by $\tau = m^{-c}$. The argument for $i \in B_*$ is identical. □

Corollary 4 (Rate)**.** *If γ, c_δ, η, κ and κ_{ref} are bounded as m grows, then $p = \Theta(\log m/m)$ suffices, and the expected number of observed entries is $\Theta(m \log m)$ — a vanishing fraction $\Theta(\log m/m)$ of the $\binom{m}{2}$ pairwise distances.*

Remark 7 (Claim discipline). (21) is a sufficient condition. No matching lower bound is established here, so Θ in Corollary 4 names the growth order of a sufficient rate, not the optimal one.

Remark 8 (The single-stage rule). Stage one alone — the raw sign rule on $\hat{u}^{(1)}$, with no refinement — is covered by Proposition 1, which gives an error rate $O(\kappa^2/(c_\delta\gamma)^2 pm)$, and by Remark 6, which gives exact recovery at the constant rate $p \gtrsim \kappa^2/(c_\delta\gamma)^2$.

Remark 9 (Entrywise alternative). The factor m lost in Remark 6 is an artefact of routing an ℓ_∞ event through an ℓ_2 bound. A leave-one-out decomposition of $\hat{u}^{(1)}$ bounds $\left\|\hat{u}^{(1)} - u^{(1)}\right\|_\infty$ directly [1, 10] and yields exact recovery from the single-stage rule at the rate of (21). The two-stage argument above is preferred here because Lemma 8 is elementary and its hypotheses are visible. Two-stage refinement of a spectral initialiser is the standard device for converting an ℓ_2 guarantee into exact labelling [15, 23].

8 What the operator selects

Theorem 1 contains a structural statement about *which* split the operator favours, independent of any sampling.

Proposition 2 (Balance preference)**.** *For a fixed edge length ℓ, the weight $w_e = 2\ell a_e b_e/m$ attached to a split of sizes (a_e, b_e) satisfies*

$$w_e = \frac{\ell m}{2}\cdot\frac{4\eta_e}{(1+\eta_e)^2}, \qquad \eta_e := \frac{\max(a_e,b_e)}{\min(a_e,b_e)}, \tag{22}$$

which is maximal at $\eta_e = 1$ and decays like $4/\eta_e$ as $\eta_e \to \infty$. An edge must therefore be longer by a factor $\asymp \eta_e/4$ to compete with a balanced edge of the same length.

Proof. Substitute $a_e = m/(1+\eta_e)$, $b_e = m\eta_e/(1+\eta_e)$ into $2\ell a_e b_e/m$. □

Two consequences are visible in simulation. First, the splits returned by the sign rule are markedly more balanced than those returned by a Fiedler vector of a similarity Laplacian, which carries no analogue of the factor $a_e b_e$. Across 1000 trees per cohort at $m = 1000$, with no subsampling and each operator using the rounding rule its own analysis states:

	Kingman		birth–death	
	$L(S)$	$\mathcal{B}$	$L(S)$	$\mathcal{B}$
valid single-edge split	996	986	996	555
median η	3.00	1.85	3.08	1.53
maximum η	499	24.6	999	5.3

The median and maximum imbalance of $\mathcal{B}$ are bounded where those of $L(S)$ are not, which is (22). The same pattern appears on random weighted trees generated without a coalescent: over 200 trees at $m = 64$ with independent uniform edge lengths, $\mathcal{B}$ returned a valid single-edge split in 200 cases out of 200 — Corollary 3 — with median $\eta = 1.29$ and maximum $\eta = 4.8$.

Second, on those same 200 trees the selected split was the one of largest weight w_e in only 40% of cases, the median rank being 2. This is Lemma 4 at work: the directions $\hat{e}_e$ are mutually correlated, so the top eigenvector of $\mathcal{M}$ is not the single heaviest projector but the direction that best explains the whole family.

The birth–death column reads the same way through Assumption D1. Under that cohort the products $\ell_e a_e b_e$ are more nearly equal across edges, Δ is small relative to w_*, and $u^{(1)}$ is a mixture of several $\hat{e}_e$ whose sign pattern need not be any single split — the regime in which Assumption D1 is the operative hypothesis and in which (21) degrades through γ.

9 Scope, and the two routes side by side

The results above are stated for additive metrics of a weighted tree, for the pairwise Bernoulli observation model (12), and under Assumptions D1 to D3 with bounded $\kappa, \kappa_{\mathrm{ref}}, \gamma, c_\delta, \eta$. Within that scope the conclusion is exact recovery of the fully observed operator's own partition, at $p = \Theta(\log m/m)$, from $\Theta(m \log m)$ observed distances.

Corollary 3 is stronger in one respect and weaker in another than the quantitative results: it holds for every weighted tree with no hypothesis at all, and it certifies that the partition is a split of T — but it says which split only through Definition 1, i.e. by naming the edge the exact operator selects. That is the appropriate form of the claim, because the distance operator and a similarity Laplacian select different edges by construction (Proposition 2), and a guarantee that pinned the distance route to the other operator's edge would be false.

The four steps correspond to those of the similarity route as follows.

Step	Similarity route	Distance route	Where they differ
1. Population geometry	block structure of L	Theorems 1 and 2	validity is unconditional here
2. Operator-norm error	matrix Bernstein	Lemma 6	symmetric-ensemble bound, no $\sqrt{\log m}$
3. Eigenvector control	resolvent expansion	Lemma 7	Davis–Kahan; no series to truncate
4. Per-node concentration	scalar Bernstein	Lemma 8	same tool, applied after refinement

Step 3 is the deliberate divergence. A resolvent expansion produces an ℓ_∞ bound in one stroke and so needs no second stage, at the price of controlling the tail of the series, which requires the ratio $\|Z\|_2/\Delta$ to be small. Lemma 7 asks nothing of that ratio and pushes the ℓ_∞ work into Lemma 8, where it is a scalar Bernstein bound over the $\asymp pm$ entries actually observed in one row.

Remark 10 (Reproducibility of the numerical statements). Every identity asserted without a displayed proof constant — (3), (4), (6), (7), (8) and (10) — was checked numerically on random weighted trees to a maximum absolute error of 1.4×10^{-14}, as was the landmark table of Section 4 and the scaling $p \propto \kappa^2 \log m/m$ of Theorem 3 over $m = 256, \ldots, 2048$.

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